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Low-cost, data-efficient, on-device soft sensors for sewer flow monitoring—learning from adjacent water level sensors

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ABSTRACT

Flow measurements are critical for sewer monitoring, but direct measurements with flow meters are often expensive due to high sensor costs and frequent sensor maintenance. Soft sensors that derive flow rates from water depth measurements are a more cost-effective approach; however, the training of such sensors still requires extensive direct flow measurements. In this paper, we propose on-device soft flow sensors based on water depth measurements at two adjacent manholes, rather than a single manhole, to reduce the demand for flow data for training. Three model structures, namely the Saint-Venant equations (SVE), a multilayer perceptron (MLP), and a physics-informed neural network (PINN), are used to implement soft sensors for two real-life pipes and one simulated pipe. In all cases, the SVE- and MLP-based soft sensors reliably estimate flow rates with a low computational load that can be implemented on a Raspberry Pi 5 that powers a water level sensor. In contrast, the PINN-based soft sensor failed due to its high computational demand. The SVE-based sensor requires much less flow data for training, while the MLP-based soft sensor delivers more accurate flow estimates but requires more flow data. Both sensors are robust against noise and bias associated with the water depth and flow rate measurements, suitable for real-life applications. The SVE-based sensor is preferrable when scarce flow data are available.

1. Introduction

As vital urban infrastructure, sewer systems collect and transport stormwater to natural water bodies and wastewater to treatment facilities (Butler et al., 2018). System failures can lead to sewer overflow and urban flooding, posing significant risks such as property damage, environmental harm, and public health hazards (Fu et al., 2008; Ge et al., 2024b; Lin et al., 2020). A key requirement for effective sewer operation and management is real-time monitoring of flow rates (Butler et al., 2018; Yuan et al., 2019; Zhang et al., 2021), which are often measured at manholes with ultrasonic Doppler flow meters (Bartos and Kerkez, 2021; Bonakdari and Zinatizadeh, 2011). However, it is challenging to deploy large numbers of flow meters on large scale networks due to their high costs and low durability (Tian et al., 2023; Tomperi et al., 2023).

As an alternative, soft(ware) sensors have been developed to estimate a variable that is difficult or expensive to measure based on easy-to-measure variable(s) (Fortuna et al., 2007). For sewer systems, a soft flow sensor could utilize water depth as the input variable, since water level sensors are usually much cheaper and more durable than flow meters (Tian et al., 2023), owing to their simple measurement principle and non-contact measurement approach (Baker, 2011; Lin et al., 2025). A water level sensor can cost <100 USD and can continuously operate for multiple years without significant degradation (Ge et al., 2024b; Tomperi et al., 2023), while measurements with flow meters can significantly drift even in the first month (ISO, 2022; Li et al., 2021; Sun et al., 2021).

For structures like weir, flume, and Combined Sewer Overflow (CSO), a simple stage-discharge relation can often be found as the model for a soft flow sensor (Fach et al., 2009; Isel et al., 2013; Kouyi et al., 2005; Mate Marin et al., 2018). For example, Ahm et al. (2016)

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developed soft flow sensors for a real-life CSO in Aarhus, Denmark, powered by empirical equations calibrated using data generated with a CFD model for this CSO. These soft sensors were validated using data collected from ultrasonic water level sensors and an electromagnetic flow meter, achieving mean relative errors of <3 % for most events. Mate Marin et al. (2018) proposed a special channel called DSM-flux that creates special hydraulic conditions suitable for establishing stage-discharge relations. This method achieved mean relative errors no >7 % for most scenarios on an experimental prototype.

Considering that water depths are often measured at manholes in sewer systems, soft flow sensors based on water depth measurements may enable systemwide low-cost flow monitoring. Tomperi et al. (2023) developed a soft flow sensor by establishing a linear relationship between flow rates and water depths for a real-life manhole in Finland, achieving R^2=0.92. Given that the stage-discharge relation is in general nonlinear, non-bijective, and even multivalued owing to the complex hydrodynamics in a manhole (Braca, 2008; Gironás et al., 2010), Lin et al. (2025) proposed a soft sensor that uses a time series of water depths at a manhole as the input to a multilayer perceptron model, to capture the nonlinear and multivalued relationship. The method was validated using three real-life manholes and a simulated manhole, achieving mean relative errors <10 %.

One limitation of the above-mentioned soft sensors, which rely on water depth measurements at a single point, is their demand for a large amount of flow data for the soft sensor training. The training data need to cover a broad range of flow conditions caused by not only inherent variations in the sewage flow but also rainfalls and tides, which influence the sewer flow rate via inflow and infiltration (Ge et al., 2024a). The collection of such flow data could last for months to cover different seasons, incurring high costs.

We hypothesize that the demand for flow data could be reduced if the soft flow sensor is based on water depths at two adjacent manholes instead of a single manhole. This is because the relationship between the flow rate through a pipe and the water depths at the two ends of the pipe is likely more easily learnable and therefore less data would be needed for the training. This study aims to develop a framework for soft flow sensor development based on water depths at two adjacent manholes. Three model structures, namely the Saint-Venant equations (SVE), a multilayer perceptron (MLP), and a physics-informed neural network (PINN), are used for the soft sensor development using data collected from two real-life sites, and data generated using simulation studies. Selected soft sensors are deployed to the microprocessor of a physical water level sensor to demonstrate their potential in real-time applications.

2. Materials and methods

2.1. General methodology

In a sewer system, a water level sensor is typically mounted at the upper part of a manhole, while a flow meter is positioned in the downstream pipe, as illustrated by the left manhole in Fig. 1(a). To estimate inflow rate of the pipe, the proposed method requires an additional water level sensor at the downstream manhole. The time series of water depth measurements from both water level sensors are the input of the proposed soft flow sensors:

$$Q_1 = f\left(H_1^{i-m_1}, \dots, H_1^{i-2}, H_1^{i-1}, H_1^{i}, H_2^{i-m_2}, \dots, H_2^{i-2}, H_2^{i-1}, H_2^{i}, \Theta\right) \tag{1}$$

where Q_i $(\mathrm{m}^3/\mathrm{s})$ is the estimated inflow rate of the pipe at time i; H_1^i (m) and H_2^i (m) are the water depth in the upstream and downstream manholes, respectively, at time i; m_1 and m_2 are the length of time series used as input for upstream and downstream manholes, respectively; f is the function that maps the water depth measurements to the flow rate; Θ denotes parameters that require calibration/training. The proposed framework supports various methods to find function f. In this study, we propose to use the Saint-Venant equations, a multilayer perceptron (MLP) model, and a physics-informed neural network (PINN) as the structure for f, as shown in Fig. 1(b, c, d). The resulting soft sensors are called SVE-, MLP-, and PINN-based soft sensors, respectively.

From a physical perspective, the flow rate Q_i at time i is influenced by both the upstream and downstream water depths in the past, i.e., domain of dependence of the Navier–Stokes equations (Szymkiewicz, 2010). The temporal span of model input (T) is expected to be relatively short, since only the most recent water depth measurements have significant influences, and should be sufficiently large to provide adequate information. Based on analysis of the characteristic curves that define the domain of dependence (Szymkiewicz, 2010), T is set to 5 min for all proposed soft sensors in line with (Lin et al., 2025) with further details provided in SI. A sampling frequency of one sample per minute is selected, in alignment with previous studies (Lin et al., 2025; Tian et al., 2023), which means $m_1 = m_2 = 5$.

2.2. Soft flow sensor models

2.2.1. Sensors based on the Saint-Venant equations

The Saint-Venant equations (SVE) are a simplified form of the Navier-Stokes equations, derived from the conservation laws of mass and momentum describing the motion of water in rivers, channels, and pipes (Butler et al., 2018; Shi et al., 2025). SVE consist of a continuity Eq. (2) and a momentum Eq. (3), which can be expressed as:

$$\frac{\partial A}{\partial t} + \frac{\partial Q}{\partial x} = 0 \tag{2}$$

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + g\frac{\partial h}{\partial x} + g\left(S_f - S_0\right) = 0 \tag{3}$$

where x is distance (m); t is time (s); $Q(x,t)$ is flow rate (m^3/s); $h(x,t)$ is water depth (m); $u(x,t)$ is velocity (m/s); $A(x,t)$ is cross-sectional flow area (m^2); g is the gravitational acceleration ($\mathrm{m/s}^2$); S_0 is the bottom slope; S_f is the frictional slope, which can be calculated using Manning's formula:

$$S_f = \frac{n^2 Q|Q|}{A^2 R^{4/3}} \tag{4}$$

where n is the Manning roughness coefficient determined by calibration; $R(x,t)$ is the hydraulic radius (m). Note that $A(x,t)$ and $R(x,t)$ are also affected by sediment depth (d_s) in the pipe, as shown in Fig. S1.

With the water depths measured at the two ends of the pipe (H_1^i and H_2^i in Fig. 1(a)) as boundary conditions, the SVE can be solved to obtain the flow rate (Gironás et al., 2010), resulting in a soft sensor. However, the solver requires initial conditions $Q(x,0)$ and $h(x,0)$ and the n value, which are typically unknown. Further, a pipe may have sediments with unknown depths (d_s), which change the relationship between $h(x,t)$ and $A(x,t)$.

The influence of unknown initial condition can be mitigated by the temporal span of model input (T). Specifically, to estimate the flow rate at time i, the SVE is solved starting from $i - T$. The initial water depths are approximated using linear interpolation of the two water depth measurements from manholes. The initial flow rates and velocities are approximated using Manning's formula (Eq (4)). Although such assumptions inevitably introduce errors, the deviation will diminish over time and eventually become negligible at time i, as Q_i is expected to be outside of the region of influence of the initial conditions (see discussion in SI).

The first-order Godunov scheme with HLL Reimann solver was used to solve SVE (detailed in SI), as shown in Fig. 1(c), which is capable of handling sharp changes and complex geometry with high accuracy (Chen et al., 2021; León et al., 2006; Yu et al., 2020). The spatio-temporal domain is discretized with Δx equaling 1 m and $CFL \leq 0.8$. The solver was implemented using Python. The unknown roughness and sediment depth are calibrated using the L-BFGS-B algorithm (Ding et al., 2004).

2.2.2. Sensors based on multilayer perceptron

Multilayer perceptron (MLP) is a widely used machine learning method and is often simple yet effective at finding hidden relationships from data (Bisong, 2019). MLP consists of at least three layers of neurons: an input layer, a hidden layer, and an output layer (Alpaydin, 2020), as shown in Fig. 1(b). The performance of an MLP model is evaluated by a loss function, which quantifies the discrepancy between observation and model prediction. Parameters ($\boldsymbol{W}_{i-1,i}$ and $b_{i-1,i}$) are trained by a backpropagation algorithm according to the gradient of loss calculated by automatic differentiation.

As established in Section 2.1, the MLP used in this study has 12 neurons (i.e., $m_1 + m_2 + 2$) in the input layer representing water depth measurements and one neuron in the output layer representing the estimated flow rate (Q_i). The hidden part consists of one hidden layer with 128 neurons, as suggested by Lin et al. (2025) and further analyzed in Table S1, providing a balance between complexity and accuracy. All layers used ReLU (Winkler and Le, 2017) as the activation function. Parameters were initialized using the Kaiming Initialization (He et al., 2015), then tuned with the Adam optimizer (Kingma and Ba, 2014) with learning rate equaling 10^{-3} and batch size equaling 1024. The loss function was Mean Squared Error (MSE), shown in SI. The MLP was implemented using PyTorch, a widely used AI framework (Paszke et al., 2019).

2.2.3. Sensors based physics-informed neural network

Physics-informed neural network (PINN) was developed to address the shortcomings of data-driven methods, especially data inefficiency and poor interpretability, by incorporating physical laws governing the system into the neural network (Raissi et al., 2019). For a system governed by 1D SVE, the PINN has two neurons in its input layer representing distance (x) and time (t) and two neurons in its output layer representing velocity $u(x,\ t)$ and water depth $h(x,\ t)$, as shown in Fig. 1 (d). The partial derivatives are obtained using the automatic differentiation (Li et al., 2024). The discrepancy between underlying physical laws and the pattern learned by PINN is quantified as residuals by substituting model outputs and partial derivatives into Eq. (2). By including the residuals (L_f) in the loss function (L), PINN can learn 1D SVE through training:

$$L_f(\theta) = \frac{1}{N_f}\sum_{i=1}^{N_f} r(x_i, t_i, \theta)^2$$

where N_f denotes the number of collocation points used to learn SVE; $r(x_i, t_i, \theta)$ is the residuals of SVE; θ is tunable parameters, i.e., $\boldsymbol{W}_{i-1,i}$ and $b_{i-1,i}$ in Eq. (4) for the vanilla PINN (Raissi et al., 2019) and hydraulic parameters n and d_s. In addition, the PINN incorporates water depth measurements at two manholes as another term (L_u) in the loss function (L):

$$L_u(\theta) = \frac{1}{N_u}\sum_{i=1}^{N_u} \left(h(x_i,\ t_i, \theta) - \widehat{h}(x_i,\ t_i, \theta)\right)^2$$

where N_u denotes the number of data points from observations; $h(x_i,\ t_i, \theta)$ and $\widehat{h}(x_i,\ t_i, \theta)$ are water depths from observations and PINN outputs, respectively. In summary, the loss function of PINN is:

$$L(\theta) = L_f(\theta) + L_u(\theta)$$

PINN could eventually learn the physical laws beneath the data by minimizing $L(\theta)$ during model training (Cuomo et al., 2022; Raissi et al., 2019, 2020).

The PINN used in this study is a 6-layer fully connected neural network with 50 neurons in each layer and a hyperbolic tangent activation function (Li et al., 2024; Luo et al., 2024; Raissi et al., 2019; Wang et al., 2021), trained with the Adam optimizer (Kingma and Ba, 2014) with learning rate equaling 10^{-3}, as suggested by previous studies (Li et al., 2024; Luo et al., 2024; Raissi et al., 2019; Wang et al., 2021). It was also implemented using PyTorch

2.3. Data acquisition

Two real-life sewer pipes and one simulated pipe were used to support sensor development. The selected real-life sewer pipes (Case Study 1 and Case Study 2) have significantly different hydraulic characteristics and are in two urbanized areas in China. For all pipes, both the upstream and downstream manholes have flow meters (model THWater TWQ) and water level sensors (model THWater TWC) measuring flow rates and water depths at an interval of 1 min.

Data from each pipe was cleaned (see SI for details) and divided into training set, validation set, and test set following various ratios representing abundant data scenarios and limited data scenarios, as will be further described below.

2.4. Real-life case studies

Case Study 1. The pipe has a diameter of 600 mm, a slope of 0.005, and a length of 70 m. The data includes both dry weather flows and wet weather flows and covers 38 days with one record per minute (Fig. S2). A total of 52,026 samples were collected, while the rest 4.9 % of records are missing. Each record consists of three values: the pipe's inflow rate and the corresponding water depths at the upstream and downstream manholes at the recorded moment. The average inflow rate of the pipe is around 210 m^3/d in dry weather and up to ~5 times higher in wet weather. It should be noted that the flows in the downstream manhole were dominated by external inflows as the downstream flow rates are multiple times larger than the flow from the upstream manhole, as shown in Fig. S2(A1) and Fig. S2(B1).

Case Study 2. The sewer pipe is a circular pipe with a diameter of 1200 mm, a slope of 0.005, and a length of 113 m. The monitoring lasted two weeks with one record per minute (Fig. S3), collecting 20,075 samples (0.4 % of samples are missing). The average flow rate into the pipe was around 16,505 m^3/d and the average water depths for upstream and downstream manholes were 0.403 m and 0.389 m, respectively.

Data separation. According to the data split convention, 70 %, 20 %, and 10 % data was used in both cases for model training, validation and testing, respectively, with details shown in Fig. S4. In addition, different portions of the Case 1 data were used for model development (training plus validation) to evaluate data efficiency of the proposed methods. These datasets ranged in size from 1-hour (~0.1 %) to 35 days (~90 %). The remaining data were used for testing.

2.5. Simulated cases

The proposed methods were also applied to a virtual pipe with various properties (roughness and sediment depths), simulated using Saint-Venant equations (SVE). These cases were included to further test the proposed soft sensors and evaluate the influences of different roughness, sediment depths, and measurement errors on the sensor performance.

The virtual pipe has geometrical properties (diameter, length and slope) identical to the sewer pipe in Case Study 1. The time series of the flow rate into the pipe and downstream water depths collected in Case Study 1 were selected as the boundary conditions. The solver introduced in Section 2.2.1 was used for data generation. The roughness varied between 0.015 and 0.04 and sediment depth varied from 0 % to 80 % of the pipe radius, resulting in 20 scenarios each with a duration of 38 days (i.e., a total of 65,664,000 data points). The method for Case Study 1 and 2 was used for data separation, i.e., 7:2:1 for training, validation, and test sets.

The simulated data had no measurement errors, while practical measurements often contain noises, biases, and drifting, which may challenge the reliability of the proposed soft flow sensors. Artificial noises, biases, and drifting were added to the case with roughness equaling 0.020 and without sediment to evaluate the robustness of the soft sensor. In total, five scenarios were evaluated: water depth measurements with noises, flow rate measurements with noises, biased upstream water depth measurements, biased flow rate measurements, and drifting flow rate measurements, as further described below.

According to specifications of commercial products from THWater, WTRExpert, Endress+Hauser, Xylem, to name a few, the errors associated with a water level sensor typically follow the Gaussian distribution with no bias, with a 95 % confidence interval of 3 mm, i.e., with mean (μ) and standard deviation (σ) equalling 0 and 1.5 mm, respectively. In contrast, the noises of a flow meter are often quantified as relative errors. The corresponding Gaussian noises had a 95 % confidence interval of 2 %, namely μ and σ being 0 and 1 %, respectively. In addition, biased errors of $+5$ mm and $+\ 0.001\ \mathrm{m^3/s}$, respectively, were added to the water depth and flow rate measurements. Drifting is commonly observed for flow meters due to their direct contact with wastewater, causing sensor fouling as well as wear and tear, to name a few. To simulate this issue, a $-5 \times 10^{-5}\ \mathrm{m^3/s{\cdot}d^{-1}}$ drift was added to flow rate data, causing $-0.002\ \mathrm{m^3/s}$ bias in the end.

2.6. Hardware

The proposed soft sensor framework includes two water level sensors, illustrated in Fig. 1(a). Both water level sensors are connected to a Raspberry Pi 5 (Fig. S5). The second water level sensor and Raspberry Pi 5 installed at the downstream manhole measures the water level and transmits data to the first Raspberry Pi 5 located at the upstream manhole, which also receives water level data from the first water level sensor (Fig. 1a). The upstream Raspberry Pi 5 runs the soft sensor algorithms described in Section 2.2 in real time to estimate flow rates. The Raspberry Pi 5 (Fig. S5) is equipped with a 64-bit Arm Cortex-A76 microprocessor, capable of executing lightweight scripts with low power consumption. LoRa, enabled through an SX1268 433 M HAT module, was used for communication between the upstream and downstream Raspberry Pi 5. Note that only the application phase was implemented on the device. The training for all soft sensors was conducted on a desktop PC with Intel Core i7–13,700 K and Nvidia 4070.

3. Results and discussion

3.1. Flow rates estimation

Real-life case studies. For Case Study 1, Fig. 2 depicts the flow rates estimated with the proposed MLP- and SVE-based soft sensors in comparison to the actual measurements. The MLP-based sensor achieved an excellent agreement between the estimated and measured flow rates, without significant under- or over-estimation, indicating satisfactory performance of the proposed soft sensor ($R^2 = 0.945$). A good fit ($R^2 = 0.931$) was also achieved by the SVE-based sensor with n and d_s estimated to be 0.0249 ± 0.0008 and 5.2 ± 0.9 mm (95 % confidence interval), respectively, although the fit was slightly inferior to the MLP-based sensor, especially during wet weather events as highlighted in Fig. 2(B1 and B2). Similar results were obtained in Case Study 2 (Fig. 3). The MLP-based sensor produced more accurate estimates of the flow rate than the SVE-based sensor, with the R^2 value being 0.914 and 0.754, respectively. Parameters n and d_s were estimated to be 0.0441 ± 0.0020 and 31.9 ± 1.2 mm, respectively. The MLP-based sensor outperformed the SVE-based sensor especially during periods with lower flows. A summary of the prediction errors is given in Table 1.

The training of the MLP-based sensor was much more efficient than the training of SVE-based sensor, taking 15.5 min and 16.4 h, respectively. This could primarily be attributed to the mathematical complexity of the SVE calibration, which needs a large amount of numerical approximation to estimate the gradient of loss for each optimization step. In contrast, MLP computes the gradient of loss analytically using a much faster method, i.e., automatic differentiation. When the trained models were deployed to the Raspberry Pi 5 device, the MLP- and SVE-based sensors produced each flow estimate in, on average, 0.0007 and 1.5 s, respectively. Both are suitable for real-time applications. The MLP-based sensor is far more efficient than the SVE-based sensor since each estimation is basically a matrix multiplication operation of a 12×128 and a 128×1 matrices plus a ReLU operation, and it was implemented using the well-optimized PyTorch package (Paszke et al., 2019). The performance of the SVE-based sensor could be further improved if a high-performance language, e.g., Fortran and *C*++, is used to replace Python, which is a general-purpose language.

Different from the SVE- and MLP-based sensors, which are trained offline with the trained model deployed on-line for real-time applications, the PINN-based soft sensor does not have a separate training step. At each time step, the model needs to be trained, which simultaneously produces an estimated flow. We trained the PINN model (Fig. 1d) on the same machine used for the training of the MLP- and SVE-based sensors (see 2.4 for details), and allowed 60 s (the sampling interval) for the training at each step. In all cases, the training was terminated due to the time constraint (rather than by the performance-based termination criteria). Consequently, the PINN-based sensor produced flow estimates that fit poorly with the measured flow rates, with R^2 values being -1.334 and - 0.324, MRE being 97.58 % and 163.18 %, and RMSE being 0.56633 and 0.43528, in Case 1 and Case 2, respectively (Table 1). The results indicate that PINN is less efficient than classic PDE solvers for a well-posed PDE problem in terms of computation efficiency and convergence. Similar findings have been reported in several other recent studies (Ferrer-Sánchez et al., 2024; Jiao et al., 2024; Ko and Park, 2025; Wang et al., 2022). Considering these unsatisfactory performances in both case studies, the PINN-based soft sensor was deemed infeasible, and not investigated further in this study.

Simulated cases. The simulated data of the virtual pipe were used to explore the effectiveness of proposed MLP- and SVE-based soft sensors under a range of pipe conditions. The pipe roughness was varied between 0.015 and 0.04 and the sediment depth varied between 0 % and 80 % of the pipe radius, resulting in 20 scenarios.

In all cases, the training of the SVE-based sensor returned estimated n and d_s values that are statistically identical to the true parameter values applied (Fig. S6). As can be seen in Table 2, the SVE-based sensor delivered nearly perfect estimates of the flow rates in all roughness and sediments conditions. As an example, Fig. 4a shows the fit between the estimated and actual flow rates in the case $n = 0.02$ and $d_s = 0$ mm (see Fig. S7 to Fig. S25 for all results). The performance of the SVE-based sensor is superior to that in the real-life case studies (Fig. 2 and Table 1) because the hydraulics in the real-life pipe may not be perfectly described by the Saint-Venant equations that underpins the SVE-based soft sensor.

As in real-life cases, the MLP-based soft sensor continues to perform well, with MRE in all cases below 3 % (Table 2), confirming that the MLP-based method has a strong ability to learn from data.

3.2. Data efficiency

In the previous section, we used around 90 % of the data to train and validate the SVE- and MLP-based soft sensors and achieved satisfactory results. Considering that the flow data are expensive to collect due to the intensive maintenance efforts required for the flow meters, there is a strong incentive to develop the sensors with a minimum amount of data. Here, we systematically varied the duration of flow data of Case Study 1 from 3 h (180 records) to 35 days (50,400 records) for sensor development, and assessed performance of the trained SVE- and MLP-based

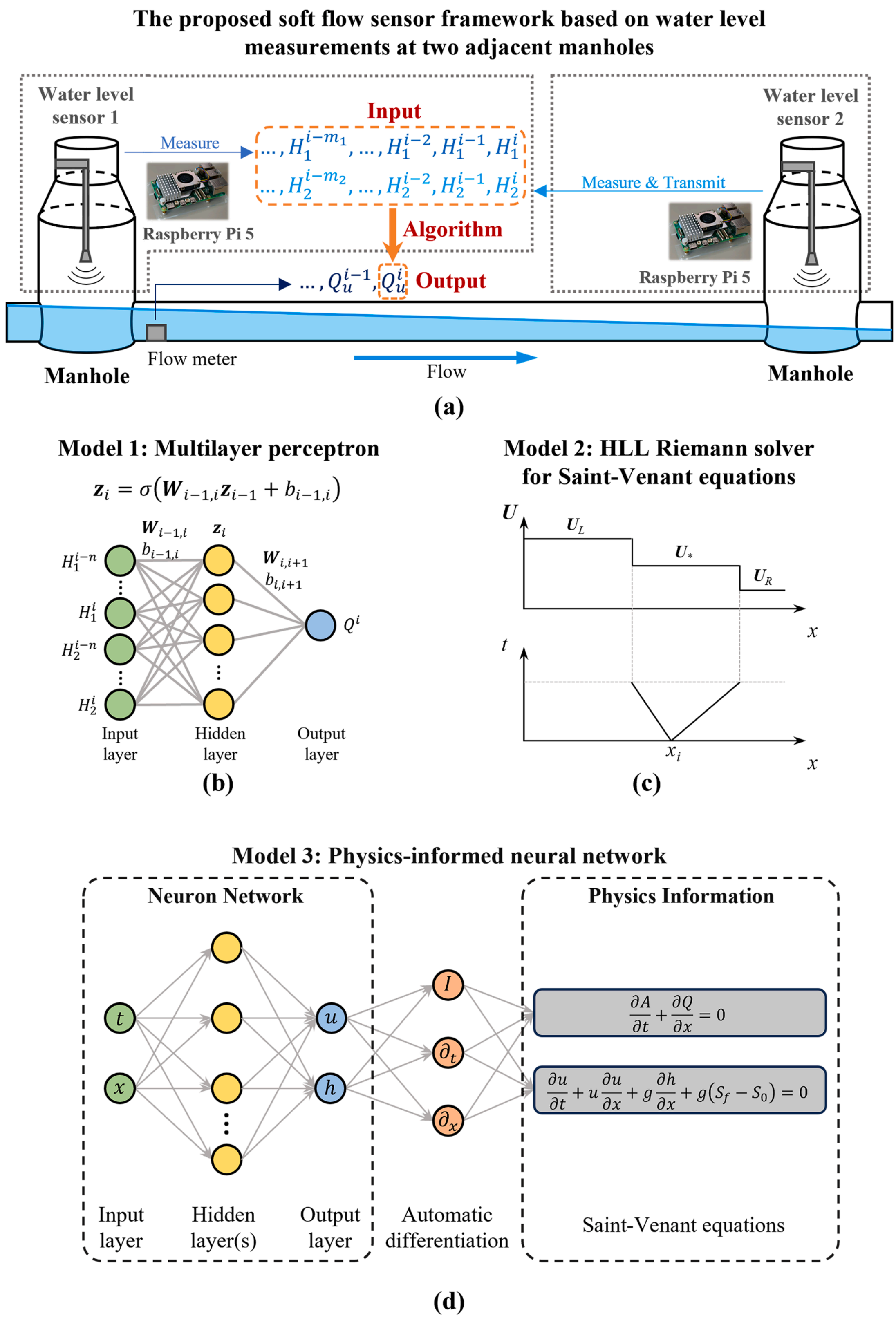

Fig. 1. Schematic of the proposed soft sensors that estimate flow rate based on two time series of water depth measurements (a), powered by: multilayer perceptron (b), Saint-Venant equations with the HLL (Harten, Lax and van Leer) Riemann solver (c), and physics-informed neural network (d). is the output of the th layer; is activation function; and are tunable weights and bias.

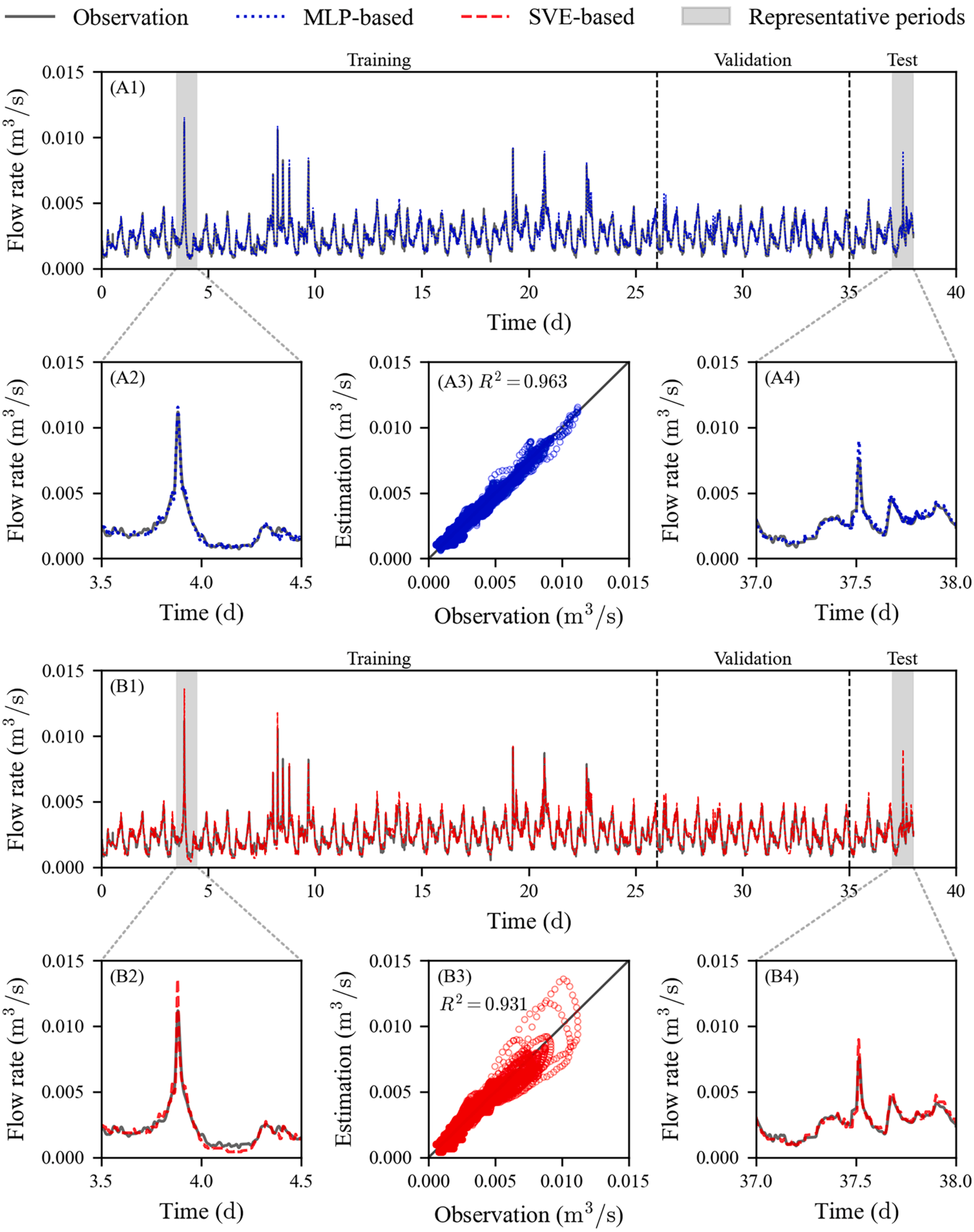

Fig. 2. Performance of the MLP-based and SVE-based soft sensors in Case Study 1: the entire dataset (A1 and B1); zoom-in views of two wet weather periods for the MLP-based (A2 and A4) and SVE-based (B2 and B4) soft sensors; comparison of estimated flow rates using the MLP-based (A3) and SVE-based (B3) soft sensors with the measured flow rate.

sensors, represented as MRE, using the wet weather flows during Days 36–38. Case Study 2 was not discussed here since it only includes dry weather flows.

As shown in Fig. 5, the SVE-based sensor is highly data-efficient achieving an MRE of 14.7 % even when trained on data with 3-hour duration (180 measurements), which was further reduced to 13.0 % when trained with 1-day data. It is worth noting that the 1-day dataset used in sensor development (training and validation) did not include any wet weather flows, indicating that the SVE-based sensor does not require wet weather flow data for the training. This is not surprising as the SVE-based sensor is supported by the fundamental knowledge on hydraulics captured in the Saint-Venant equations. These equations only

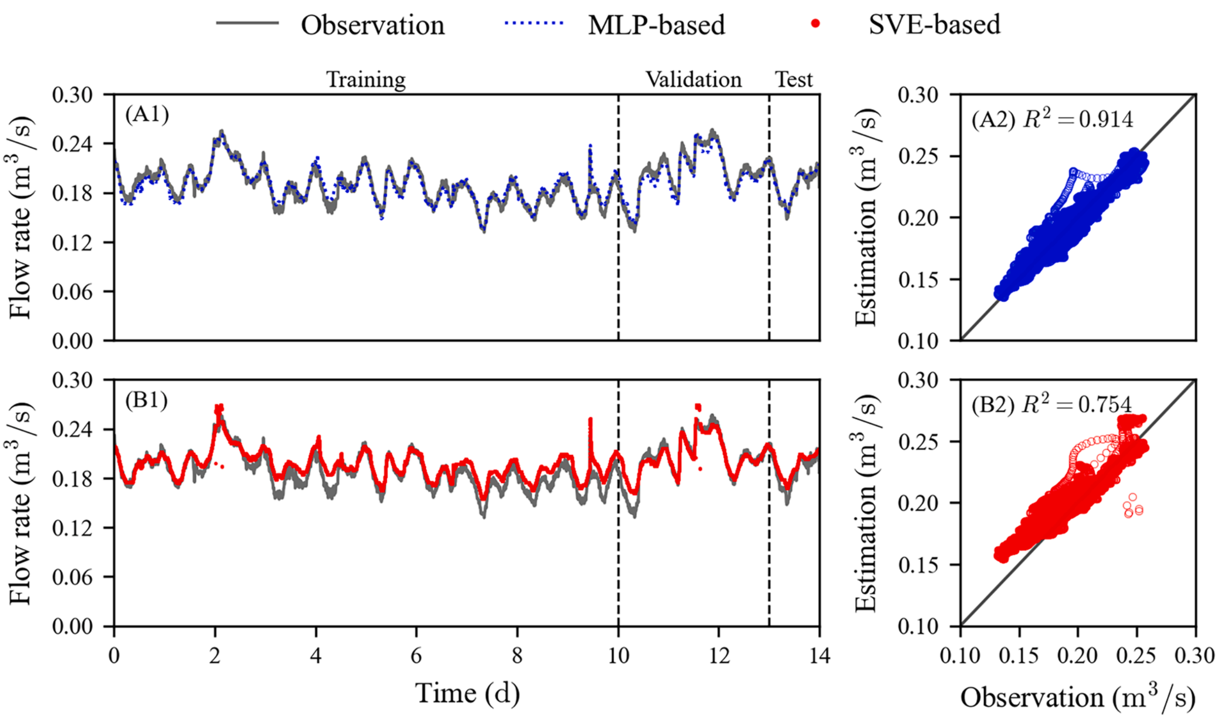

Fig. 3. Performance of the MLP-based and SVE-based soft sensors in Case Study 2: the entire dataset (A1 and B1); comparison of estimated and measured flow rates using the MLP-based (A2) and SVE-based (B2) soft sensors.

Table 1
Performance of the three soft flow sensors in the two case studies.

	Case	Case Study 1			Case Study 2		
	Sensor	R^2	MRE	RMSE	R^2	MRE	RMSE
1	SVE-based	0.931	9.03 %	0.00028	0.754	5.11 %	0.01165
2	MLP-based	0.963	7.27 %	0.00020	0.914	2.94 %	0.00691
3	PINN-based	−1.334	97.58 %	0.56633	−0.324	163.18 %	0.43528

have two parameters, namely n and d_s, to be learnt, which does not require extensive datasets. Further increasing the dataset size from 1 day to 35 days only led to marginal improvement, with the MRE value further decreasing to 11.8 %. As discussed in the previous section, the SVE-based sensor would have prediction errors when the flow data cannot be fully described by the Saint-Venant equations.

In contrast, the MLP-based sensor had an MRE of 54.7 % when trained with 3-hour data (Fig. 5). The MRE progressively decreased with increases in the size of the training dataset, reaching 5.60 % when the data size increased to 11 days. The two rain events in the 11-day dataset each led to a sharp decrease in the MRE value (Fig. 5), suggesting that, for the MLP-based sensor, it is important to have wet weather flows in the training datasets. This is as expected as, unlike the SVE-based sensor, the MLP-based sensor does not have any fundamental hydraulic knowledge built into the model structure and hence requires the training data to capture the relationship between the flow rate and the water levels under a broad range of conditions. This is also a common shortcoming of data-driven models. Compared to the SVE-based sensor, the MLP-based sensor produced smaller MRE values (5.09 % vs. 11.8 %) when an adequate amount of data (35 days) is used for training. This is because the MLP model has better learnability due to its flexible model structure, while the SVE model is more rigid comprising two learned parameters only. Therefore, a hybrid approach that employs SVE to capture the fundamental dynamics and uses MLP (or other data-driven models) to further learn from the residuals could be promising. Alternatively, the Saint-Venant equations could be augmented, with additional model components and parameters, to account for the differences between the measured data and the estimated flow with the standard SVE.

3.3. Robustness against measurement errors

The simulated data of the virtual pipe was used to analyze the sensitivity of the soft sensors to measurement noises and biases, tested with wet weather flows in the test set. Without measurement errors, the soft sensors were able to accurately estimate flow rates, as shown in Fig. 4. By adding Gaussian noise to the flow rate measurements, the MRE slightly increased from <0.01 % to 0.10 % and from 0.73 % to 1.07 % for SVE- and MLP-based sensors, respectively, indicating that both soft sensors are insensitive to random measurement errors. When the

Table 2
Performance of MLP- and SVE-based soft sensors for simulated scenarios with various roughness and sediment depth in terms of mean relative error.

Roughness	Mean relative error (MLP)					Mean relative error (SVE)				
	Sediment depth					Sediment depth				
	0 %	20 %	40 %	60 %	80 %	0 %	20 %	40 %	60 %	80 %
0.015	1.36 %	1.99 %	2.24 %	2.48 %	2.67 %	<0.01 %	<0.01 %	<0.01 %	<0.01 %	<0.01 %
0.02	1.28 %	1.92 %	2.24 %	2.39 %	2.42 %	<0.01 %	<0.01 %	<0.01 %	<0.01 %	<0.01 %
0.03	0.98 %	1.62 %	2.07 %	2.11 %	2.20 %	<0.01 %	<0.01 %	<0.01 %	<0.01 %	<0.01 %
0.04	1.18 %	1.52 %	1.86 %	2.20 %	1.81 %	<0.01 %	<0.01 %	<0.01 %	<0.01 %	<0.01 %

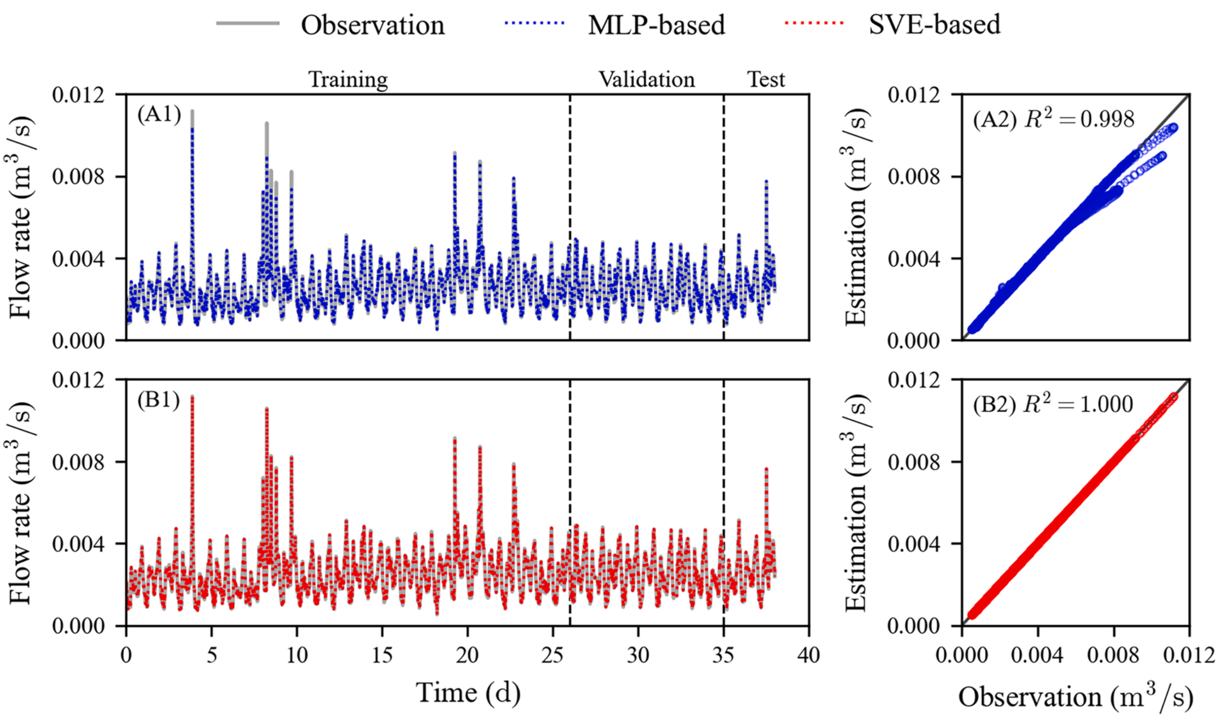

Fig. 4. Performance of the MLP-based and SVE-based soft sensors in an error-free simulated case (and mm): the entire dataset (A1 and B1); comparison of estimated and observed flow rates using the MLP-based (A2) and SVE-based (B2) soft sensors.

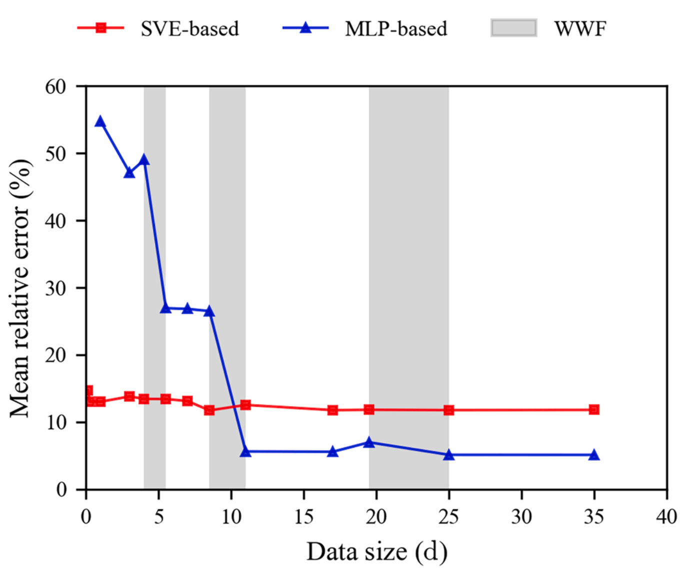

Fig. 5. Performance of the SVE- and MLP-based soft sensors for wet weather flows with different data size for model development.

Gaussian noise was added to the water depth data used for training, the MRE increased from <0.01 % to 4.86 % and from 0.73 % to 2.43 % for the SVE- and MLP-based sensors, respectively. The error growth is much smaller than the magnitude of the noise, which is 7.66 % of mean water depth. Note that the MLP-based sensor was less affected by noise than the SVE-based sensor, likely due to MLP's strong ability to generalize patterns from noisy data (Alpaydin, 2020). Adding artificial noises to training data, regarded as data augmentation, is a common trick to improve MLP's performance (Rebuffi et al., 2021). In summary, these testing results suggest that the soft sensors are robust to random measurement noises in both the water depth and flow rate measurements. Given that a 1-min sampling frequency is sufficient to develop an effective soft sensor, the robustness could be further enhanced by increasing measurement frequency and incorporating an online noise reduction filter.

By using an upstream water level sensor with a +5 mm bias (12.8 % of the mean water depth), the SVE- and MLP-based soft sensors achieved MRE of 7.92 % and 0.54 %, respectively. This indicates that the MLP-based sensor was capable of learning from the biased data, while the SVE-based could reduce the impact of bias on the output. We also added a bias of $+0.001$ m^3/s to the flow data used in the training, which represents an MRE of 40.2 % in the model input. This resulted in an MRE of 43.7 % and 17.7 % for the SVE- and MLP-based sensors, respectively, indicating both methods did not amplify errors in the flow measurements.

The introduction of $a\ -\ 5\times\ 10^{-5}$ $m^3/s{\cdot}d^{-1}$drift to the flow rate measurement (Fig. S27(A)) caused a significant decline in the performance of the MLP-based sensor trained and validated with 2-week data, with an MRE of 17.6 %. In comparison, the SVE-based sensor, trained using 1-day data, experienced only a slight negative impact, resulting in an MRE of 1.07 % (Fig. S27(B)). This reflects an important advantage of the SVE-based sensor, which does not require flow measurement over an extended period.

The SVE- and MLP-based soft sensors have also been applied to raw measurement data, as shown in Fig. S26. The MLP-based soft sensor maintained satisfactory performance, achieving a R^2 of 0.878. It adequately captured peak flow events and automatically smoothed out noise in base flows. However, the SVE-based soft sensor failed in this case, since noise in the raw data produced physically implausible values, e.g., inflow depth is (or very close to) zero while inflow rate is nonzero, resulting in an infinitely high inflow velocity. Therefore, real-time noise smoothing is necessary, especially for the SVE-based soft sensor.

3.4. *Choice of sensors*

Compared with Doppler flow meters, soft flow sensors are potentially more cost-effective because a water level sensor is substantially cheaper and requires less maintenance costs (Li et al., 2021; Tomperi et al., 2023). Three soft sensors, powered by SVE, MLP, and PINN, respectively, were investigated in this work. The SVE- and MLP-based soft sensors were demonstrated to achieve reliable estimation of sewer flow rates in both real-life and simulated scenarios. Both sensors are robust against errors associated with water depth and flow measurements, and the SVE-based sensor is also robust against flow meter drifting. Both sensors are computationally efficient, and their algorithms can be implemented on-device (e.g., a Raspberry Pi 5) for real-life applications. To the contrary, the PINN-based soft sensor requires on-line rather than off-line training, which is too slow to be implemented in real-time, rendering its real-life application infeasible.

The SVE-based soft sensor is the preferred option when limited flow data are available for sensor development, particularly when the available flow data do not cover wet weather periods. The fundamental knowledge captured in the Saint-Venant equations enables the application of the trained SVE-based sensor under a wide-range of conditions including those not covered by the training data.

While requiring more data for development, the MLP-based sensor achieves more accurate estimations of the flow rates compared with the SVE-based sensors, especially when the training data sufficiently cover wet weather flows. With no prior knowledge built into the model structure, the MLP-based sensor learns the relationships between flow rates and water depths purely from data without any assumption. This makes the sensor more flexible without being influenced by any mismatch between assumptions and reality. The MLP-based soft sensor is a preferred option when it is feasible and affordable to install flow sensors over a long-term (e.g. months) that cover both wet and dry weather conditions. Besides, while a specific SVE-based soft sensor has to be trained for each individual site, an MLP-based sensor could potentially be trained using data from multiple sites and consequently the same sensor could be deployed to multiple sites. Parameters such as diameters, slopes, roughness, and sediment depths will likely be needed as model inputs in such cases.

In-sewer conditions may undergo progressive changes (e.g. sediments can build up over years), and hence both types of soft sensors would require regular retraining to be able to reliably estimate flow under changed conditions. The more data-efficient SVE-based sensor is advantageous as a short (e.g. a few hours) deployment of the flow sensor is adequate to provide new data for the fine tuning of the model.

4. Conclusions

A soft sensor framework for estimating sewer flow rate based on water depth measurements at two adjacent manholes was proposed. Three sensor structures were designed and implemented. The following conclusions are drawn:

- Both the Saint-Venant equations and a multilayer perceptron are suitable model structures for estimating sewer flow rates based on water depth measurements at two adjacent manholes. In contrast, a physics-informed neural network is not a suitable model structure for a soft flow sensor.
- Time series of water depths at both manholes, each with a 5-min duration, provide sufficient inputs for the soft flow sensors, independent of the model-structure adopted. This aligns with physical reasoning, since the head difference between two water level sensors defines the pressure gradient that drives the flow.
- Both the SVE- and MLP-based soft sensors are computationally efficient, and their algorithms can be directly deployed to an on-site computing device, e.g. one embedded in a water level sensor.
- An SVE-based soft sensor is data-efficient, achieving satisfactory performance with 1-day data (or even a few hours).
- In comparison, an MLP-based soft sensor requires more data for development, which should last for e.g. weeks covering both dry and wet weather conditions, but can be more accurate than an SVE-based soft sensor.
- Both the SVE- and the MLP-based sensors are robust against measurement noises and biases associated with the water level sensor and flow meter.

CRediT authorship contribution statement

Ruozhou Lin: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Wenchong Tian:** Writing – review & editing, Validation, Resources, Methodology, Conceptualization. **Ruihong Qiu:** Supervision, Methodology. **Lihan Hu:** Methodology. **Zhiguo Yuan:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare a personal relationship which may be considered as potential competing interests: an author of this submission, Zhiguo Yuan, serves as Editor-in-Chief of Water Research X. The manuscript has been handled and reviewed independently by other editors of the journal, with Zhiguo Yuan recusing himself from any decisions related to the evaluation and acceptance of this work.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.wroa.2025.100415.

Data availability

The data that has been used is confidential.

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